

Detection of Emotions from Speech using Deep Learning Techniques and Traditional Techniques: A Survey

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Abstract—Detection of emotions from speech data is an important but exigent task of Human-Computer Communication. Speech Emotion Recognition system is a hybrid model that analyses and detect emotions embedded in speech signals. Emotion detection is a prominent task as emotions are instinctive in nature. Recently, various technologies are used to extract emotions from speech databases including traditional approaches for Speech Emotion Recognition (SER). As a substitute to traditional techniques in SER, Deep Learning techniques is great weapon to deal with the challenges in SER. This paper presents a brief review on various Deep Learning techniques and throws light on some latest literature relating to speech-based emotion recognition system.

Keywords—Speech emotion recognition, Deep learning techniques, Deep Boltzmann machine, Deep Reinforcement Learning, Deep recurrent neural network, Deep belief network, Convolution neural network

I. INTRODUCTION

Voicing of sentiments are processed by various measures of human body signs such as language, deportment, human gesticulation, acknowledgement, attitude and motion. SER is the mechanism to let machine recognize sound and process them. When machine interpret the speech and translate them to readable text then it is known as automatic speech recognition. Emotions can be expressed in humans by anatomic processes like rate of heart, skin reflection, body thermal reading, tissue and neural membrane potentials etc. These parameters can easily predict emotional state of speaker without any physical contact. Recognition of emotions from speech could be affected by some elements like ‘what is spoken’ (element 1), ‘how it is presented’ (element 2) and ‘who speaker says it’ (element 3). In the first element, input speech tells about linguistic speech which relies on the pronunciation of words, and the next element is depended on user’s emotion state so it is called as paralinguistic information and the last third element deals with Accumulative information which consist of speaker’s primary features such as speaker’s age, sex, and body dimension [1].

As it is well known, additional messages are conveyed by speech beyond the words, such as sentiments or distinguishing speaker. Emotion recognition is growing on a wider way with the huge success of human-computer interaction. There

abundance of benefits for both user and machine if Machine will be able to successfully recognize emotion from the speech. However, this is interesting but challenging task as well because expression of emotion varies from one person to another. Emotion detection from speech is evolved from HCI [2]. To avail the facility of natural communication with machine direct voice commands are given rather than of using traditional devices. Some systems use dialogue systems for speech recognition for example BPO, vehicle driving system, medical applications.

Speech emotion recognition is the novel approach to recognize different sentiments embedded in humans. In SER speech signal acts as the input to the machine which is further processed to extract emotions from the voice data. The basic goal of SER is to develop a model which will automatically recognize emotions from the speech signal. SER has indeed garnered significant attention within the research community due to its wide-ranging applications in various fields. SER involves the identification and analysis of emotions conveyed through speech signals, and its applications span from human-computer interaction to mental health monitoring and beyond.

HCI system are not recommended for speech recognition because it consist of many flaws. Thus, to overcome these problems effective efforts should be made to enhance and achieve better accuracy in SER. Feature extraction and features classification are primarily two phases used in SER. features classification can be done by the usage of non-linear/linear classifiers.

Bayesian Networks (BN) or the Maximum Likelihood Principle (MLP) and Support Vector Machine (SVM) are mostly used as linear classifiers for emotion recognition. Usually non-linear classifiers give good performance in speech emotion classification because speech is non-linear itself. Gaussian Mixture Model (GMM) and Hidden Markov Model (HMM) are some well known non-linear classifiers [3]. Some of the most commonly used classifiers for emotion detection are Mel-Frequency Cepstrum Coefficients (MFCC), Perceptual Linear Prediction cepstrum coefficients (PLP), Linear Predictor Coefficients (LPC), Mel Energy-spectrum Dynamic Coefficients (MEDC), including K-Nearest Neighbour (KNN) [4].

Machine Learning includes various technology like Deep learning, Artificial Intelligence etc, out of that deep learning has got more attention and emerged in the field of SER. The deep learning technique trains the computer model to classify the data directly from images, text, sound, natural language processing, and pattern recognition. Deep Neural Networks consist of feed-forward network which has input layer, output layer, and multiple hidden layers. There are many feed-forward network like Deep Neural Networks (DNNs), and Convolution Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM). CNN provides better results in image and video classification where

RNNs, LSTM provides better result in the field of SER and as natural language processing [5].

II. LITERATURE REVIEW

S.NO	References	Work Done	Approach/ Classifier	Results
1	Siddique Latif et.al (2020) [2]	Presented semi-supervised AE based multi-task framework which improved the performance in SER. IEMOCAP and MSP-IMPROV datasets are tested on proposed model.	Adversarial Autoencoder(AAE), CNN, 10-fold cross-validation technique	68.8 % accuracy achieved for IEMOCAP dataset and 63.6 accuracy achieved for MSP-IMPROV dataset.
2	J. Schmidhuber et al (2015) [5]	This report presented a detail description on Deep learning neural network including supervised, un-supervised network, LSTM, RNN, CNN, DBN, DBM etc	MFCC, LPC, MEDC, PLP, KNN, LSTM, RNN, CNN, GPU-CNN, GPU-MPCNN, AE, HMM, HRL, RL,SOTA,TDNN, WTA etc.	The presented report shows detailed theory about every deep learning neural network and compared them successfully
3	S. Demircan et. al (2014) [7]	The presented model successfully classified emotions from speech data using KNN. Author used MFCC for feature extraction.	MFCC, K-NN	Designed model extracted features in reduced form by MFCC. At the beginning data size was 24933×1. Later on size decreased as 388×16, and Classification accuracy is found %50
4	F. Dellaert et. al (1996) [8]	The presented model Classified emotions from speech data. It aims to select efficient features for SER and to implement the same author used various techniques for feature extraction and classification.	MLB classifier, KR, KNN, PFS, FS, Selective composition, Cooperative composition	The presented model tested emotional speech with over 1000 utterances from different speaker and classified from MLB, KR, KNN where MLB error rate is 44%, KR error rate is 35 %, KNN error rate is 32%. Thus, concluded to use KNN for classification, for feature selection promising first (PFS) and forward (FS) feature selection is tested and compared and FS technique gave better performance as PFS leaves 8 attribute while FS leaves only 5 attribute.
5	A. D. Dileep et. al (2013) [9]	The presented model, proposed HMM based IMK for detecting sequential patterns of diverse length. Later performance of the SVM, HMM-based IMK is tested for E-set in English alphabet and	HMM, GMM, SVM, IMK	98.43 % accuracy is obtained for SVM based classifier with LMHIMK(large margin hidden 1) using the complete sequence matching Method. The classification accuracy is seen about 4.5% higher than the conventional MLHMM

		identification of consonant, vowel in Hindi language.		classifier for E-set recognition task.
6	O. Pierre-Yves et. al (2002) [10]	The author compared a large set of neural networks, SVM, KNN, decision trees, together with 200 features, using a large database. Various features sets are tested on different machine learning algorithms, and efficiency is calculated.	1-NN, 5-NN, 10-NN, Decision Trees/C4.5, Decision Rules/PART, Kernel density, K star, Linear regression, LWR, Voted perceptron's, SVM degree 1, SVM degree 2, SVM degree 3, SVM 4, VFI etc	Results comparison- 1-NN 85.1%, 5-NN 78.9%, 10-NN 69.4%, Decision Trees/C4.5 79.1%, Decision Rules/PART 80.1% Kernel density 84.2%, Kstar 75.6 %, Linear regression 74.8 %, LWR 79.1%, Voted perceptrons 50.2%, SVM degree1 83.2%, SVM degree2 85.4%, SVM degree3 84.9%, SVM 4 85.1 %, VFI 77.1%, M5Prime 80.9 %, Naive Bayes 85.1%, AdaBoostM1 84.2 %, AdaBoost M1/PART 90
7	Virendra Chauhan et. al (2016) [11]	Presented system successfully converts input speech signals into text format to help disable students. MFCC used for feature extraction and GMM is used for training audio files	MFCC, GMM, Expectation-Maximization (EM), Maximum Likelihood Estimation (ML).	Speech data is successfully converted in text form with accuracy of 71%.
8	Jinjin Y et. al (2002) [17]	The presented model successfully classified phoneme using naive bayes classifier. Author tested isolated Wcative, vowel, and nasal phonemes from TIMIT corpus database. Naive Bayes classifiers is trained on 6 male speakers and tested on 3 male speakers.	Naive Bayes classifier, Histogram	Phoneme recognition results of fricatives is 58.94%, Phoneme recognition results of vowels 33%, Phoneme recognition results of nasals is 16.67 %, Phoneme recognition results of fricatives with Normalization is 61.59 %, Phoneme recognition results of vowels with Normalization is 34.49%
9	S. K. Bhakre et. al (2016) [18]	Presented model classifies the speech signal into 4 classes i.e., anger, happy, sad, neutral. It classifies statistical features of energy, pitch, MFCC, ZCR.	Naïve Bayes Classifier, MFCC, ZCR	Accuracy achieved for anger, happy, sad, neutral is 81%, 78%, 76%, 77% respectively
10	O. Eray et. al (2018) [20]	Author presented Turkish speech recognition application consists of coding of speech and recognition of speech . Recorded words are encoded by LPC and for speech recognition Soft Margin SVM (SM-SVM) classifier and the Least Square SVM (LS-SVM) classifier are used.	LPC, LS-SVM, SM-SVM	Patterns are successfully recognized with the accuracy of 91% with SM SVM classifier during the testing while LS SVM classifier recognized with a success of 71%
11	T. Wien et. al (2016) [21]	The author aims to recognize and classify emotions from speech data with Naive Bayes Classifier. SER is done using MFCC, and pitch related features extracted from a speech segments.	Naive Bayes Classifier, MFCC, SVM, GMM, HMM, KNN	Model is tested on Open SMILE library with overall 78% of accuracy achieved.
12	S. POUYANFAR et. al (2018) [22]	The presented model gives the summary on different deep learning techniques with framework and applications.	RvNN, RNN, CNN, DBM, DBN , GAN, VAE	Presented model briefly introduces the traditional neural networks and several supervised deep learning

				algorithms, including RvNN, RNN, CNN, DBM, DBN , GAN, VAE. Moreover, several optimization techniques have also been provided.
13	R. A. Khalil et. al (2019) [23]	Presented model gives the overview of various Deep Learning techniques utilized for SER system.	DBM, RNN, DBN, CNN, AE	Presented model gives brief review on deep learning methods used for recognition of discriminant emotions like sad, angry, happy, neutral etc.
14	D. Neiberg et. al (2006) [25]	Presented model performs automatic emotion recognition by using MFCC and GMM. Acoustic features are calculated in between range of 20 to 300 Hz. Author combines MFCC, MFCC Low and GMM to provide better results.	MFCC, GMM	Acoustic 3gram classifier provides better accuracy and 80% accuracy is achieved.

Speech is one of the attracting medium which can be used to express emotions from language. So determining emotional feature from correct speech signal window is an important task. As seen and discussed in (M. Swain et.al (2018) [1]) to improve SER's system is a prominent task. Author presented a brief review on different papers published on speech emotion recognition and concluded that from 2000 to 2017 various researchers were interested to work with SER.

Majority of research outcomes concluded to use database of speech with minimal number of speakers so that user will get speaker specific information. As discussed in (S. G. Koolagudi et. Al (2012) [3]). The presented model gives more preference to the use of hybrid model for SER. GMM, AANN, or SVM will may provide good results in SER with Simulated emotional speech corpus, Elicited speech corpus, natural speech corpus, Excitation source features, Prosodic features.

SER system comprises of three phases i.e., signal processing, feature calculation and classification. Digitalisation and segmenting of input signal is involved in signal processing whereas in feature calculation relevant features are extracted from the acoustic signal with respect to emotions and lastly classification phase classify segmented speech signal into emotion classes as discussed in (T. Vogt et. al (2008) [6]). Author presented a model which gives the summary on Automatic emotion recognition from speech and compared different research paper's results as Petrushin et al (2002) provide 68 % accuracy in emotion recognition.

In this research work (K. P. GORAI et al., 2017 [12]), represented a speaker recognizing model which recognizes the silent discriminants and useful features of speech, a mix of techniques have been presented. Paper presented speaker recognition system which determines speaker's authenticity. Combination of GMM, HMM, SVM, Linear predictive cepstral coefficients (LPCC) is used to enhance accuracy in speaker recognition. Out of 630 speakers 192 are female and 438 males are successfully1 recognized.

To solve obstacles coming in Speech Recognition a triply embedded algorithm of HMM should be used. HMM algorithm are now widely used to represent phoneme like sub-words, syntax and words in advanced systems. HMM creates astronomical number of equivalent states which makes recognition of isolated and connected word quite easy as discussed in(Rabiner et. al (1989) [13]).

In (N. Najkar et. al (2010) [14]) HMM-based speech recognition systems is designed. Main core of this system Viterbi algorithm. Viterbi algorithm uses dynamic programming to get best prediction results. The presented model uses particle swarm optimization strategy to recognize speech . Author recognized isolated words and breaks consonants phones in the experiment. Presented model used MFCC, HMM, Baum–Welch algorithm, LCPSO, PSO for speech recognition with successfully recognized speech words with reduced error rate of 30.29%.

Study [24] presented literature review on different classifiers, algorithms, datasets which are probable to enhance accuracy in speech emotion recognition. It summarizes researches carried out in between 2000 and 2017 on SER. According to which bulk of researcher focuses on feature extraction algorithm which provide good features for classification and enhance accuracy

III. TRADITIONAL APPROACHES FOR SER

Traditional speech recognition method comprises of major 4 steps: 1. Building up of input signal, 2. Extraction of features, 3. Selection of features, 4. Classification of features. Data used for input is a mixed data and may consist of rough and noisy data, which results in inaccuracy of classification results. So data pre-processing is needed. Data pre-processing is the initial step in the data analysis pipeline where raw data is cleaned, transformed, and organized to make it suitable for further analysis [6]. After pre-processing of input speech signal important features are extracted and classifies the data in valuable information.



Figure 1: Traditional SER system

Classification are of two types, short term classification and long term classification. Short term classification consist of short term properties like frequency, sound intensity,toughness [7] whereas standard deviation is used as main feature in long term classification [8]. GMM, HMM, K

nearest Algorithm, SVM, Bayes Classifiers are very commonly used for SER which is specified in table 1.

Table 1: Linear/Non-Linear Classifiers

Classifiers	Linear/Non-Linear	Uses	References
GMM	Non-Linear	clustering, image segmentation	[14]
HMM	Non-Linear	speech recognition, Writing recognition	[9]
K-nearest	Linear	Classification of machinery data	[10]
Bayes Classifiers	Linear	Credit Scoring, medical data classification	[19]
SVM	Linear/Non Linear	handwriting detection, intrusion detection, face recognition, email classification	[21]

A. GAUSSIAN MIXTURE MODEL GMM is a probabilistic density function composed by the summation of all the weighted inputs of gaussian component densities. It is primary used for recognition biometric system classification .GMM is also used with the combination of MFCC to produce better results[11] . Gaussian distribution plays an important role in GMM as GMM is represented by Gaussian distribution only, which is calculated by mean, variance, and weight of Gaussian distribution. Gaussian Mixture density can be expressed as

$$p(\bar{x}|\lambda) = \sum_{i=1}^M p_i b_i(\bar{x})$$

Where $b_i(\bar{x})$ is the component densities,

and $\bar{x}$ = mean vector

The main advantage of GMM is it gets trained very fast can add new speaker in the system very easily. Designing a speaker model with GMM is produces valid spectral features [12]. GMM uses two methods to accomplish its task i.e, Maximum likelihood(ML) and maximum a posteriori (MAP). Maximum likelihood method produces best results when GMM has enough training dataset, and when training data is in-adequate MAP is used.

B. HIDDEN MARKOV MODEL (HMMs) are widely known for solving 'labeling' problems [13]. HMMs can extract hidden data from detectable sequential symbols. HMM determines the hidden parameters from the experimental parameters.Determining these hidden parameters typically involves the use of algorithms like the Baum-Welch algorithm (also known as the Expectation-Maximization algorithm for HMMs) or training methods like supervised learning where the model is trained on labeled data. These methods iteratively

adjust the parameters of the HMM to maximize the likelihood of the observed data given the model. HMM can work in numerous fields like speech recognition, character recognition, bio-informatics, statistical pattern recognition. An HMM model is also known as to be finite state machine, where every state is denoted as a single or a multi-model Gaussians mixture. HMM generally uses left-to-right topology to fulfil observation criteria [14]. Left-to-right topology checks the transition from every state to itself, and also to the neighbors. Maximum likelihood method helps to find the parameters for HMM training sets [14].

C. K NEAREST NEIGHBOR Classifier is a machine learning approach designed to do classification and prediction of data. It is non-parametric supervised approach used widely to solve complex problems. It finds the related features for classification of data using proximity approach [15]. KNN divides the data in two cases, first is the available case and second is the predictable case. The data to be tested from KNN shall given to classifier for the classification, where minimum distance between the result set and data features are calculated. The data features having minimum distance with defined classes are assigned to that defined class. For example a machine has to determine the type of vehicle. It has different vehicle image category stored in the database. For the classification input vehicle type will be compared with the stored data and the data whose feature are quite similar to the result data will be assigned to that specified category. Like if vehicle is type of car then it will go in car category, if it is truck then it will go in Truck category. The KNN does the classification by finding near-by feature around the classes. The data is classified by calculating similarity distance between the data from the defined classes.

D. BAYES CLASSIFIERS is inspired from bayes theorem . It is a supervised algorithm meant to solve classification problems. It does the classification by predicting the probability in between the object as it is a probabilistic classifier too. Bayes Classifiers has various applications like sentimental analysis, text recognition, spam filtration, classifying articles, speech recognition etc. Bayes classifier believes that every feature of certain data is independent of occurrence of other data's features [16]. In bayes classifier conditional probability is computed for every classes and the classes with maximum conditional probability are selected [17]. Automatic speech recognition requires trained datasets which consist of speech signals with their specified emotions. Bayes classifier classifies the trained datasets and when the classification is accomplished classifier will be able to predict the emotion of the data which is kept for testing of machine [18].

E. SUPPORT VECTOR MACHINE SVM is the supervised algorithm designed to address the classification and regression problem. SVM portraits hyperplanes to classify n-dimensional space data. Various n-dimensional space data are segregated into different defined classes by creating decision overlays , so that classification of upcoming new data points will become easy. SVM selects main vectors for designing hyperplane, and these hyperplane incorporates vectors hence known as support vector machine [19]. SVM uses quadratic programming approach to solve classification, regression problems. SVM has many application like speech recognition, face recognition, 3-dimensional object recognition, text recognition, handwriting recognition etc. [20]. Results of all traditional classifiers are shown below in the table

Table 2: Results comparison of different traditional classifiers

Classifier	Accuracy Achieved	References
GMM	69%	[21]
HMM	78%	[14]
KNN	70-80	[21]
Bayes Classifier	77%	[21]
SVM	70-80%	[21]

IV. DEEP NEURAL NETWORKS FOR SER

Traditional SER technique's efficiency was dependent to the type of clean data, a noisy data often led to slower performance of the machine, therefore engineers focus on feature selection first before any type of data classification during traditional approach. SER system involves various steps like feature selection, feature extraction, acoustic modeling, emotion recognition. The traditional SER mechanism involved various classifiers to accomplish the SER process like GMM was used for acoustic modeling whereas HMM was used to deal with variation of voice [1]. Thus Traditional SER approach was not much efficient as every task was accomplished from the instruction given from the researcher itself, comparatively DNN performs features extraction automatically which facilitate researcher to get desired features of data without any expertise knowledge of feature's domain [22]. Figure 2 represents a clear difference in between traditional approach and deep learning approach [23].

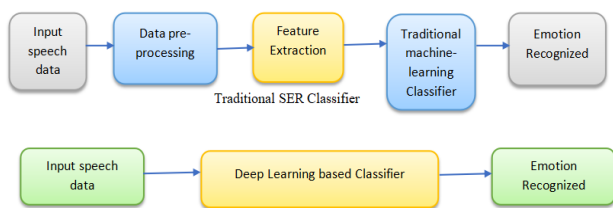


Figure 2: Deep learning-based SER classifier vs Traditional SER classifier

DNN classifiers comes from family of machine learning which is well known for Layered Architecture. Layered architecture of DNN allows machine to extract features from the lower layer and the lower layer architecture extract features from their respective sub layer and the process goes on. Thus DNN does the features extraction in automated way [22].

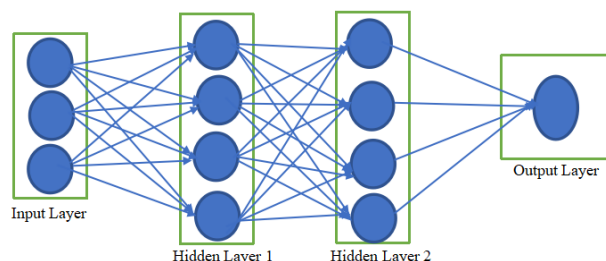


Figure 3: Deep neural network architecture

Deep learning is a multi-layered structure which can be unsupervised, semi-supervised or fully supervised represented in figure 3. It delivers results very efficiently. With the great victory of DNN in numerous fields, DNN is the most trending researched technology running now a days. This paper

represents a brief survey on different deep learning algorithm like CNNs, RNNs, DBNs, DBMs and AE which is discussed in coming sections.

A. CONVOLUTIONAL NEURAL NETWORK (CNN)

CNN is another well-known deep learning algorithm based on feed-forward architecture. CNN are extensively used for various application like pattern recognition, sentiment analysis, natural language processing, speech processing, computer vision etc. CNN are widely used for image classification and can be extended for voice / speech recognition too.

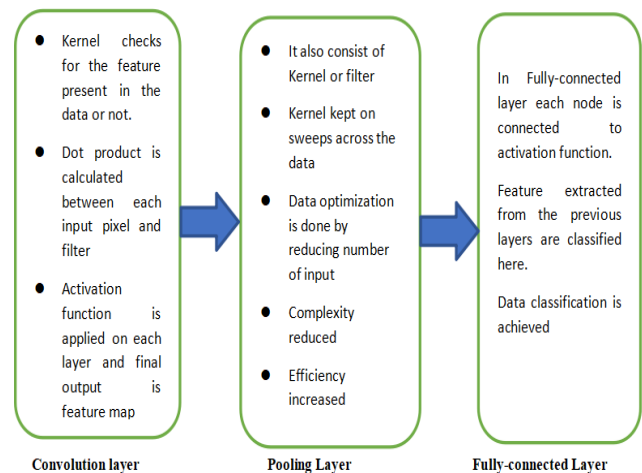


Figure 4: CNN process model

The architecture of CNN is inspired from the neuron in human brains. As it is known CNN is multi-layered architecture and each layer has neuron that process the input data and gives respective results. CNN consist of convolution layer, pooling layer, fully-connected layer which is explained in figure 4.

B. RECURRENT NEURAL NETWORK (RNN) is also known as sequential supervised neural network where input layer is interdependent on the output layer. It is feed-forward network where output from older layer is fed as an input to the next layer. RNN can predict the output from the corresponding input as it can remembers previous data, and for the same it has memory to keep previous processed data. The main problem with RNN its gradients gets disappears after some time interval, and to avoid such situations LSTM is used to data between recurrent connections. [23]

B. DEEP BELIEF NETWORK (DBN) is another Unsupervised Probabilistic Deep learning algorithm inspired from Restricted Boltzmann Machine (RBM). DBN is having collection of hidden layers and visible layers where top layers are undirected and bottom layers are directed. Hidden layers stores the features that shows relationship between the data where the visible layer holds the output. DBN follows probabilistic, statistic approach which enables it to produce every possible values. DBN gives efficient results in classification of data. DBN can provide better accuracy in speech emotion recognition as it can remember previous pattern for long duration of time. DBN has various application like data classification, image recognition, voice

recognition, data optimization etc. The pictorial representation of DBN is presented in figure 5.

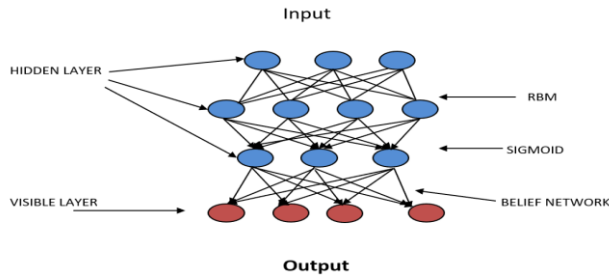


Figure 5: DBN architecture

D. DEEP BOLTZMANN MACHINE (DBM) DBMs is another deep learning model inspired from Markov Random fields. It consist of multiple hidden layers with directionless connections in between the nodes as shown in figure 5. The variable of these hidden layers are chosen randomly and embedded with stochastic entities [23].

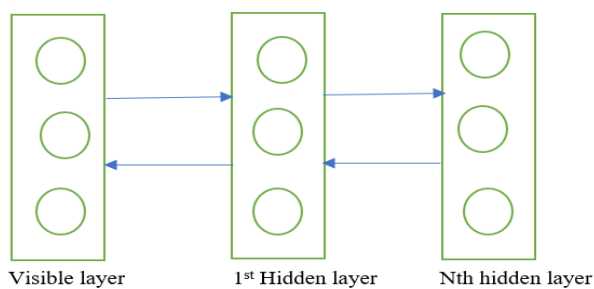


Figure 6: DBM architecture

DBM architecture is based on back-propagation network, features are trained hierarchically from the input and extracted features are sent as an input to the next layers. Advantage of DBM is its ability to learn features and gets trained very fast and provides very efficient results. DBM can give better results in SER if input data is acoustic.

E. AUTO ENCODER AE is another deep learning algorithm usually used for data optimization. AE is an unsupervised learning algorithm destined to do data compression. Auto-encoders encode data by compressing it into a latent representation, then reconstructing the original data from this representation. They consist of an encoder and a decoder. AE can rebuild the input data in new compressed form. AE comprises three layers i.e., encoder, coder, decoder. Brief representation of AE layers is shown in figure 7.

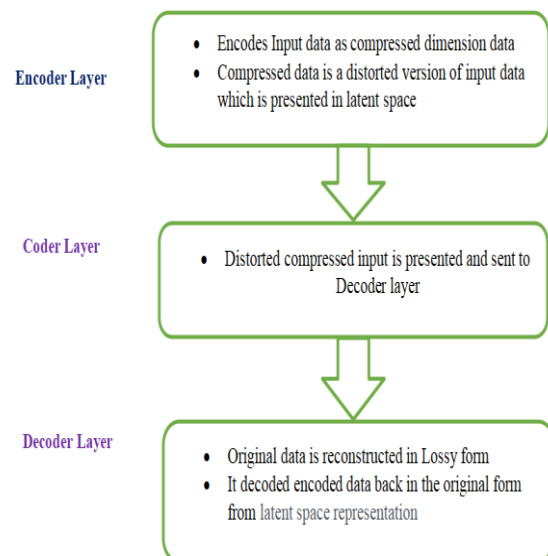


Figure 7: Auto Encoder System

Autoencoder classifiers are mainly used for image compression, data optimization etc. Any data can be reduced dimensionally using AE. The disadvantage of AE is it can remove some important features too while compression of data.

V. CONCLUSION

Emotion detection and recognition is a task which has importance in various aspects. It could be made very applicable in various areas if the whole process of emotion detection could be made accurate, speedy and applied on various types of data such as text, sound, voice, and pictures. A number of techniques falling under conventional and deep learning have been reviewed for this purpose in terms of knowing the work, approaches and their results. The traditional approaches covered are GMM, HMM, K-nearest neighbours, Bayes classifier and SVM. They altogether are good at clustering, image segmentation, speech recognition, face detection and other tasks. The accuracies of traditional techniques range from 69% to 80%. For higher accuracies deep learning techniques are reported to be better in the range of 70% to 90+%. The deep learning techniques considered perform different tasks and they are CNNs, RNNs, DBNs, DBMs and AE.

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